PassFrame tuition

Coordinates & Transformations

Year 5 Accelerated — 11+ Lesson Worksheet (PassFrame Tuition)

Worked examples — tutor talks through these first

Easy: A point A is at (-7, 4). What are its coordinates after being translated 5 right and 9 down?
Method & answer: 5 right: -7 + 5 = -2. 9 down: 4 - 9 = -5. New point: (-2, -5).

Standard: A triangle has vertices at (1, 1), (1, 5) and (4, 1). It is reflected in the y-axis. What are the new coordinates of each vertex?
Method & answer: Reflecting in the y-axis reverses the sign of x: (-1, 1), (-1, 5), (-4, 1).

Advanced: A point B is at (-4, 6). It is rotated 90° anticlockwise about the origin, using (x, y) → (-y, x), and the result is then translated 3 down. What are the final coordinates?
Method & answer: Rotate: (-y, x) = (-6, -4). Translate 3 down: (-6, -7).

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A point A is at (-8, 11). What is its x-coordinate?

2. A point A is at (-8, 11). What is its y-coordinate?

3. A point starts at (-12, 7) and is translated 15 right and 20 down. What are its new coordinates?

4. A point starts at (9, -14) and is translated 23 left and 18 up. What are its new coordinates?

5. A point is at (-15, 8). It is translated 20 right. What are its new coordinates? A) (5, 8) B) (-35, 8) C) (-15, 28) D) (-15, -12)

6. A point is at (-11, -9). Which quadrant of the grid is it in?

7. A point is at (14, -6). Which quadrant of the grid is it in?

8. A point starts at (-16, 5) and is translated 9 right and 12 down. What are its new coordinates?

9. A point starts at (7, 19) and is translated 25 left and 9 down. What are its new coordinates?

10. Select ALL that apply. Look at the points P(-6, 14), Q(6, -14), R(-6, -14) and S(6, 14). Which do NOT lie in the third quadrant? A) P(-6, 14) B) Q(6, -14) C) R(-6, -14) D) S(6, 14)

11. A point starts at (0, -17) and is translated 13 left and 21 up. What are its new coordinates?

12. A point starts at (-19, -14) and is translated 27 right and 22 up. What are its new coordinates? A) (8, 8) B) (-46, -36) C) (8, -36) D) (-46, 8)

13. A line segment joins A(-9, 3) and B(11, 3). What is the midpoint of AB?

14. A point starts at (13, -8) and is translated 21 left and 15 up. What are its new coordinates?

15. Which of these points lies exactly on the y-axis? A) (0, -13) B) (-13, 0) C) (13, 13) D) (-13, -13)

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A point is at (-14, 9). It is reflected in the x-axis. What are its new coordinates?

2. A point is at (16, -11). It is

6. A point is at (12, 9). It is reflected in the x-axis, then the result is reflected in the y-axis. What are the final coordinates?

7. A point is at (-19, -14). It is

11. A point is at (-22, -16). It is reflected in the x-axis. What are its new coordinates?

12. Select ALL that apply. A

reflected in the y-axis. What are its new coordinates?

3. A point starts at (-17, 13) and is translated 24 right and 19 down. What are its new coordinates?

4. A shape has vertices A(-6, -6), B(-6, 2), C(3, 2) and D(3, -6). It is translated 8 right and 8 up. What are the new coordinates of A, B, C and D?

5. A point at (-13, -7) is reflected in the x-axis. What are its new coordinates? A) (-13, 7) B) (13, -7) C) (13, 7) D) (-13, -7)

reflected in the y-axis. What are its new coordinates?

8. A line segment joins A(-8, -5) and B(-8, 13). What is the length of AB, and what is its midpoint?

9. A point starts at (21, -17) and is translated 29 left and 12 up. What are its new coordinates?

10. A point at (17, -12) is reflected in the y-axis. What are its new coordinates? A) (-17, -12) B) (17, 12) C) (-17, 12) D) (12, -17)

rectangle has vertices at (-9, -4), (13, -4), (13, 10) and (-9, 10). Which statements are true? A) Its width is 22 units B) Its height is 14 units C) The vertex (13, 10) is diagonally opposite (-9, -4) D) Its perimeter is 40 units

13. A point starts at (-4, -4) and is translated 25 right and 25 up. What are its new coordinates?

14. A point at (-5, 15) is reflected in the x-axis to give P, then P is translated 7 down. What are the final coordinates?

15. A point starts at (-13, 16) and is translated 20 right and 23 down. What are its new coordinates? A) (7, -7) B) (-33, -7) C) (7, 39) D) (-33, 39)

Standard / Hard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A point is at (6, 13). It is rotated 90° clockwise about the origin. What are its new coordinates?

2. A point is at (9, -8). It is rotated 90° anticlockwise about the origin. What are its new coordinates?

3. A point is at (-11, 15). It is rotated 180° about the origin. What are its new coordinates?

4. A point at (12, 5) is rotated 180° about the origin. What are its new coordinates? A) (-12, -5) B) (12, -5) C) (-12, 5) D) (5, 12)

5. A point is at (0, -18). It is rotated 90° clockwise about the origin, using (x, y) → (y, -x). What are its new coordinates?

6. A point starts at (-9, -21) and is translated 27 right and 8 up. What are its new coordinates?

7. A square has vertices at (-7, -7), (7, -7), (7, 7) and (-7, 7). What is its side length, its perimeter, and the coordinates of its centre?

8. A point is at (-8, -13). It is rotated 90° anticlockwise about the origin. What are its new coordinates?

9. A point is at (-17, 21). It is reflected in the x-axis. What are its new coordinates?

10. Select ALL that apply. A shape has vertices A(-3, 0), B(-3, 9), C(6, 9) and D(6, 0). It is rotated 90° clockwise about the origin, using (x, y) → (y, -x). Which of these are correct new vertices? A) A becomes

11. A point starts at (14, -19) and is translated 27 left and 30 up. What are its new coordinates?

12. A point at (9, -14) is reflected in the y-axis, and the result is then rotated 90° clockwise about the origin using (x, y) → (y, -x). What are the final coordinates?

13. A rectangle has vertices at (-11, -8), (5, -8), (5, 6) and (-11, 6). What is its perimeter? A) 60 units B) 30 units C) 44 units D) 76 units

14. A triangle has vertices at (2, 1), (2, 7) and (6, 1). It is translated by the vector (translation of 4 left, 3 down). What are the new coordinates of all three vertices?

15. A point starts at (-13, -13) and, after a translation, ends at (0, 0). What single translation vector describes this move?

(0, 3) B) B becomes (9, 3) C) C becomes (9, -6) D) D becomes (0, -6)